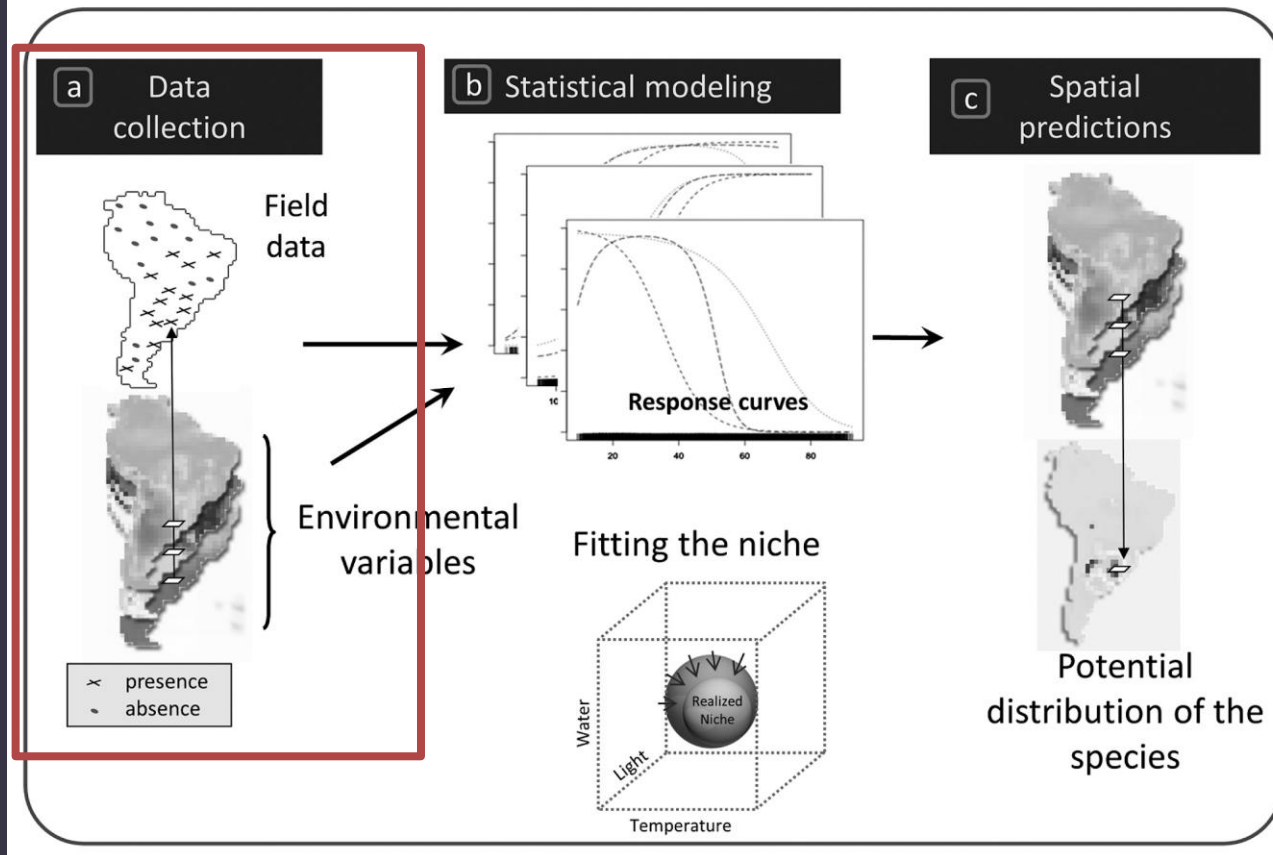


SDM Crash Course #2: Presence and Background Data



What are we doing today?

From Niche to Distribution: Basic Modeling · 43



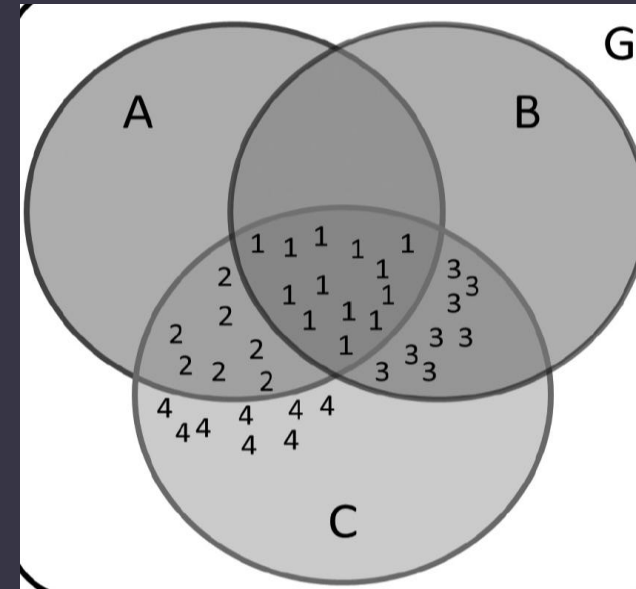
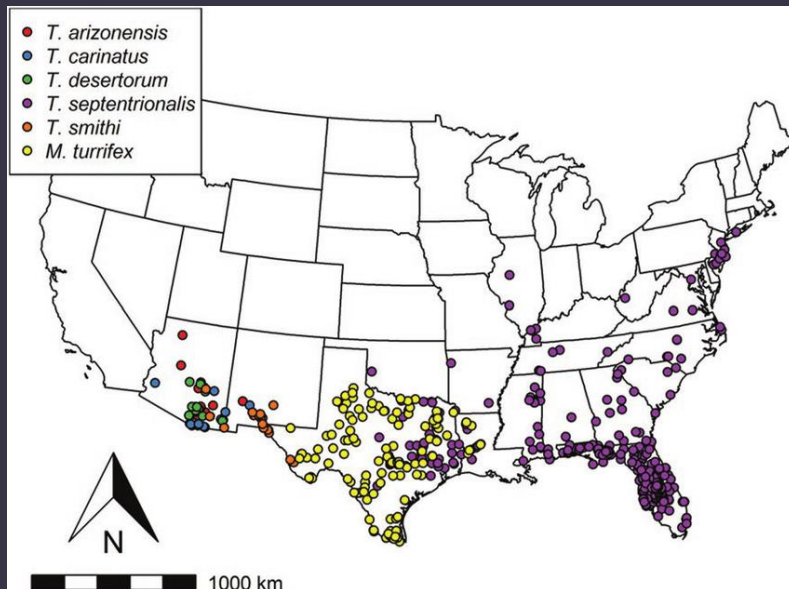
SDM Data:

- Occurrence data
 - In a perfect world
 - In the real world
- Background data
 - What is it
 - Considerations

Occurrence Data

Presence: Where a species is known to occur at a space and time

- Not *necessarily* where a species can persist in space and time
- Used to correlate species occurrence with predictor variables in correlative SDM frameworks



Properties of a Well-Behaved Sample

Random

- Probability of sampling an occurrence is proportional to the probability that the species occupies the space
- **Not biased**

Independent

- Probability of sampling an occurrence is not influenced by other occurrences
- **Not autocorrelated**

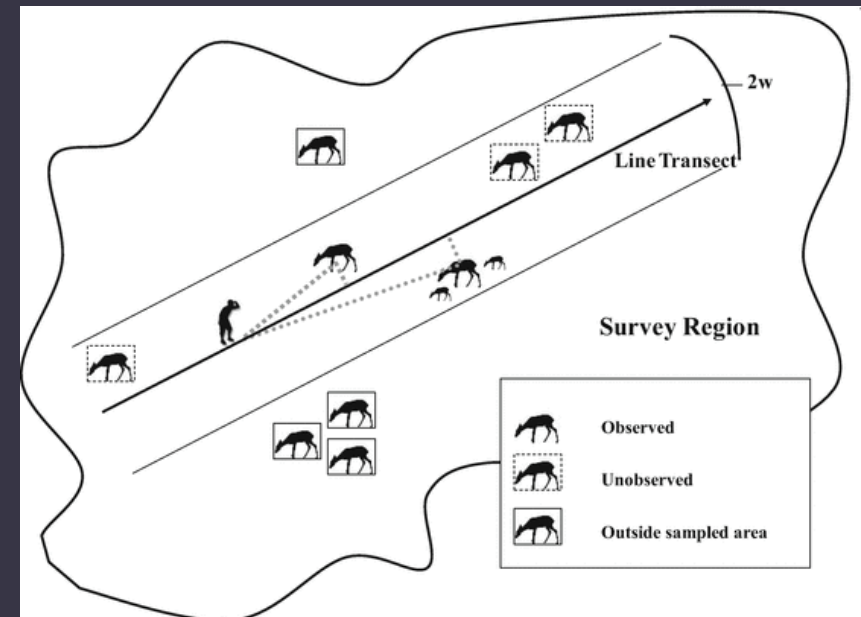
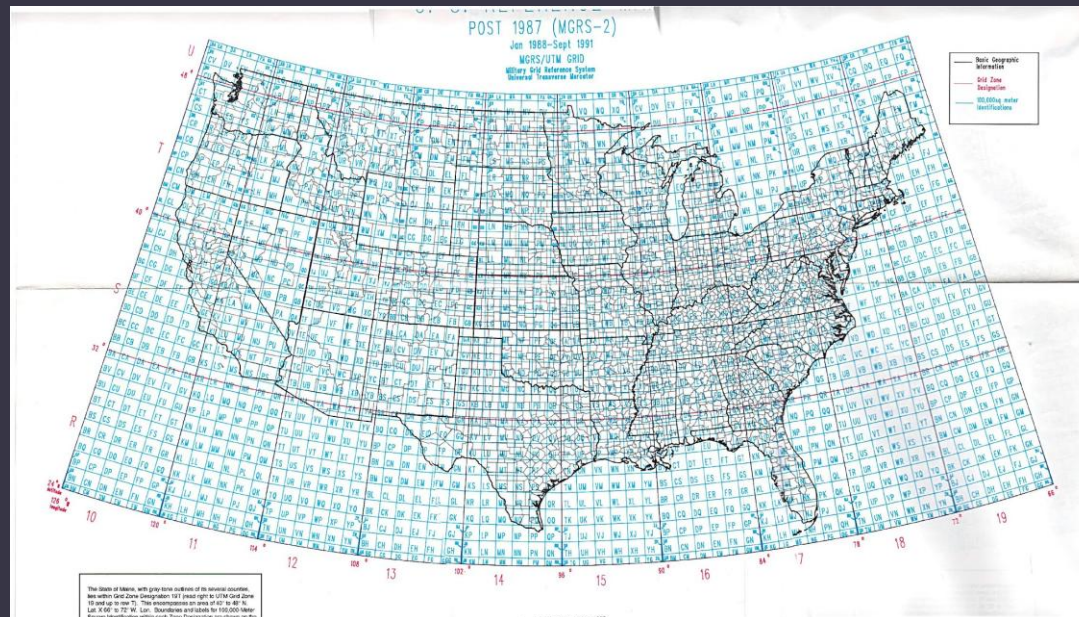
Sufficient

- Enough samples to approximate the distribution
- **Adequate power**

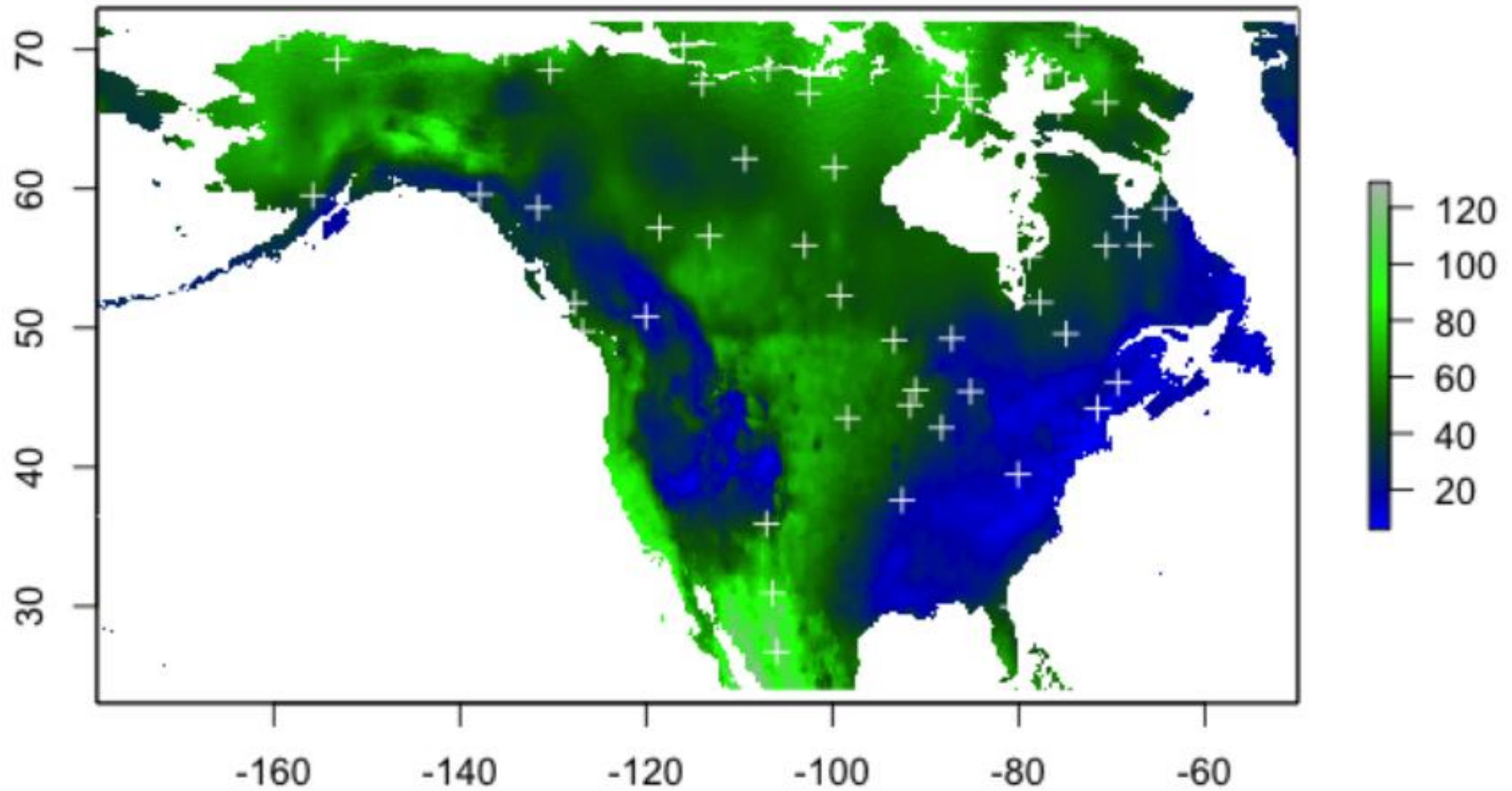
In a Perfect World: Random

Randomly sampled:

- Grid (Cartesian)
- Transect (Vector)



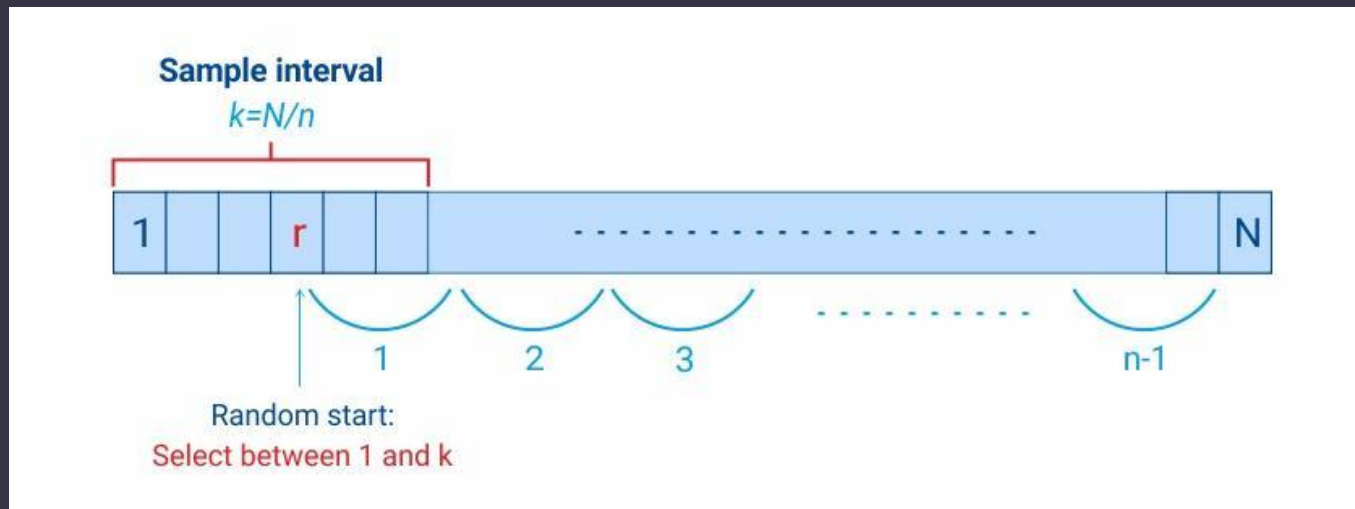
Random



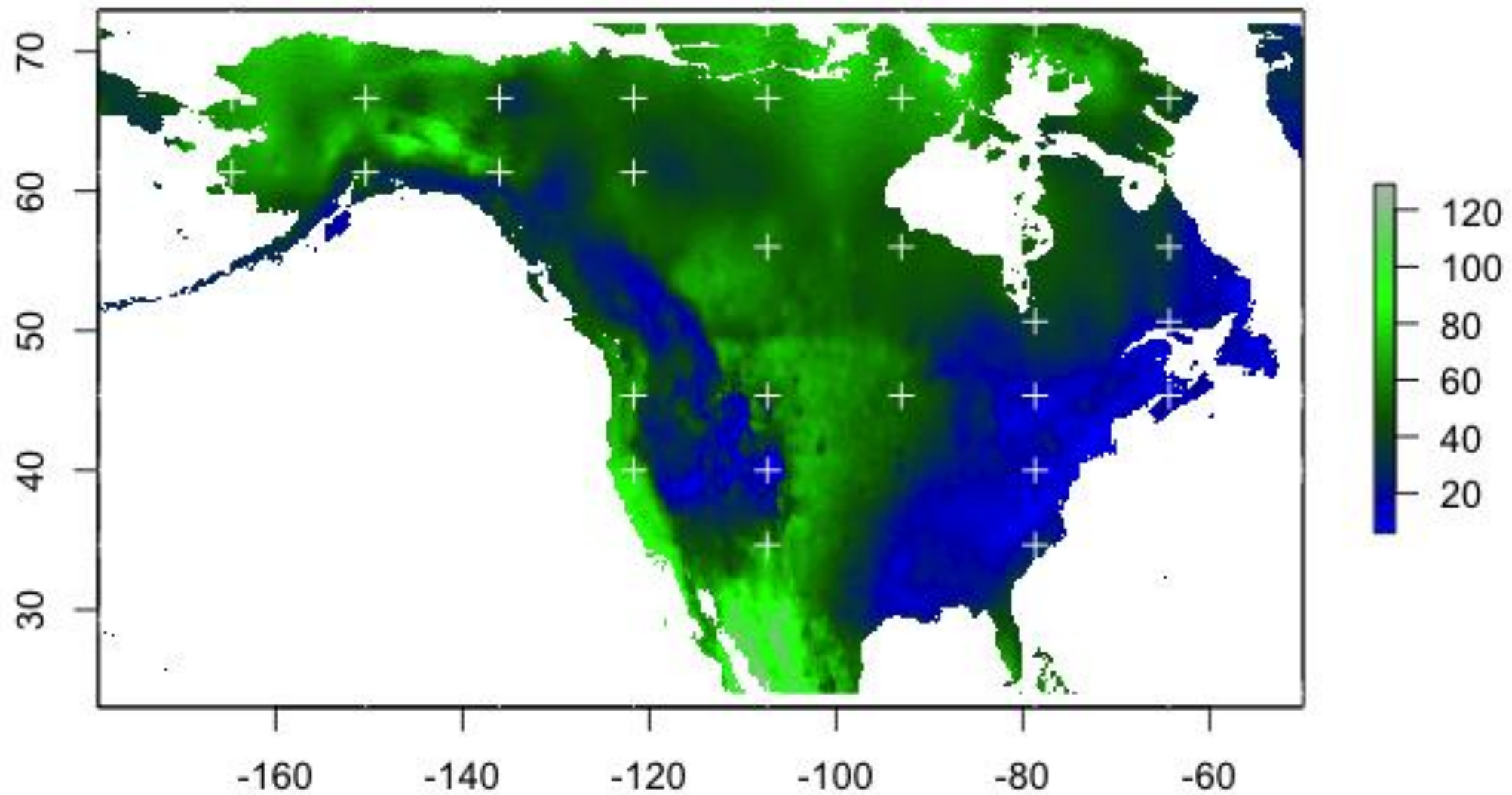
In a Perfect World: Independent

Independent Data:

- Maximize average nearest neighbor distance
- Sample at fixed intervals
- Utilize systematic or random uniform sampling



Random & Systematic

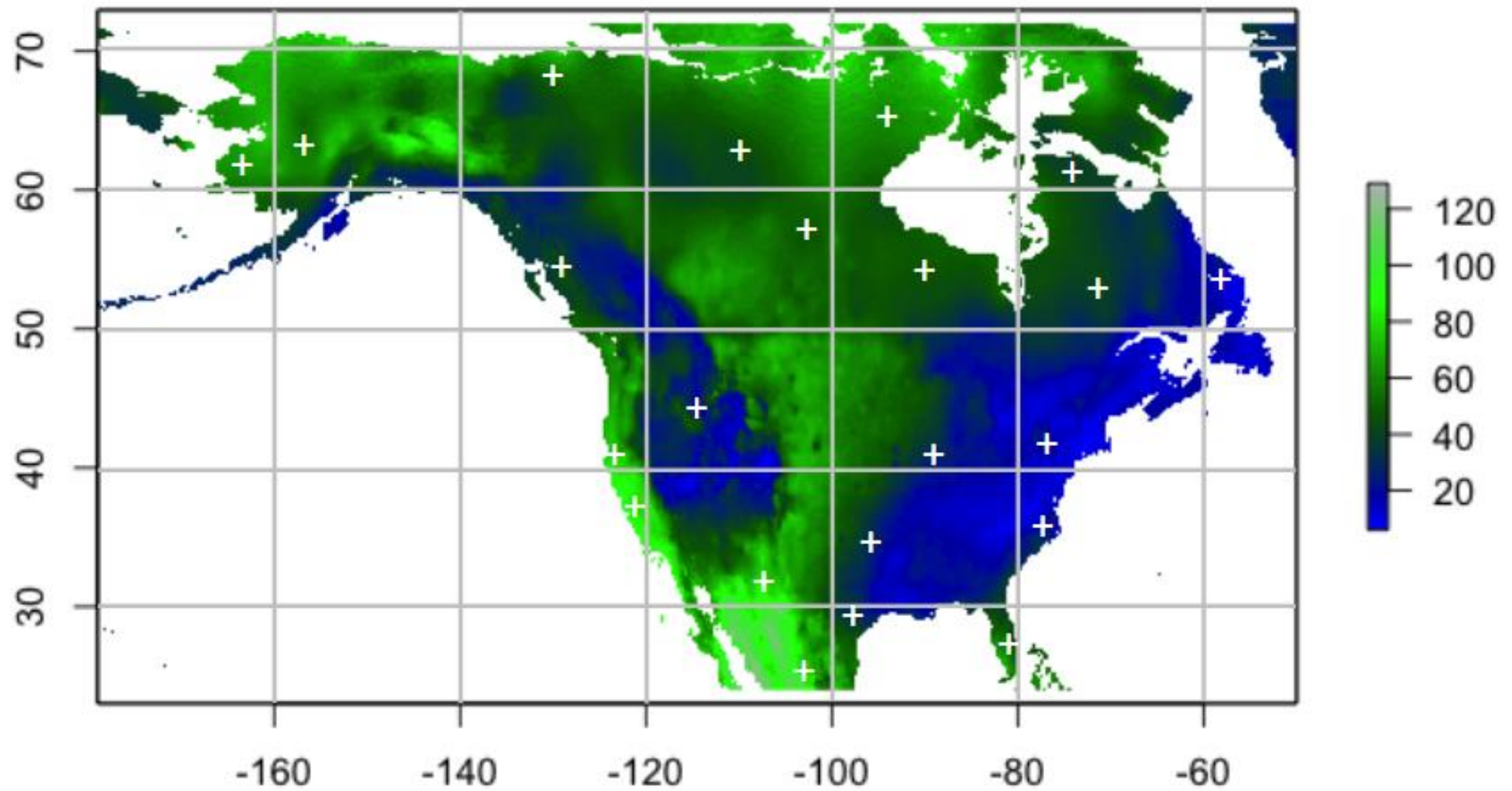


In a Perfect World: Sufficient

Sufficient Data:

- How many samples exactly will depend on the study species
- Rough rule of thumb: 10-20x the number of predictors
- Coverage: samples should be taken from all over the extent
 - Stratification: Divide samples into discrete areas, sample randomly from each to obtain coverage

Random & Stratified



Real Data is Imperfect

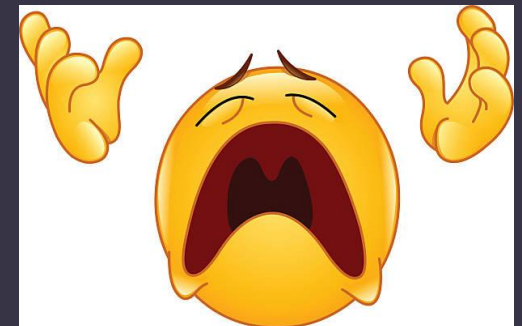
Table 7.1 *Examples of databases that store species distribution data.*

General	GBIF	www.gbif.org
General	Map of Life	https://mol.org/
General	LifeMapper	lifemapper.org/
General	IUCN Red List	www.iucnredlist.org/
Herps	HerpNET	herpnet.org/
Mammals	MaNIS	vertnet.org/
Marine species	OBIS	www.iobis.org/
Amphibians	AmphibiaWeb	http://amphibiaweb.org/
Birds	ORNIS	http://ornisnet.org
Birds	Bird Life	www.birdlife.org/
Plants	Atlas Flora Europaea	www.luomus.fi/en/ database-atlas-florae-europaeae/
Plants	BIEN	http://bien.nceas.ucsb.edu/bien/
Central America	REMIB	www.conabio.gob.mx/remib_ingles/ doctos/remibnodosdb.html?
Brazil	SpeciesLink	http://splink.cria.org.br/

Screenshot

Collections & online repositories are used more than perfectly designed surveys

Often opportunistic



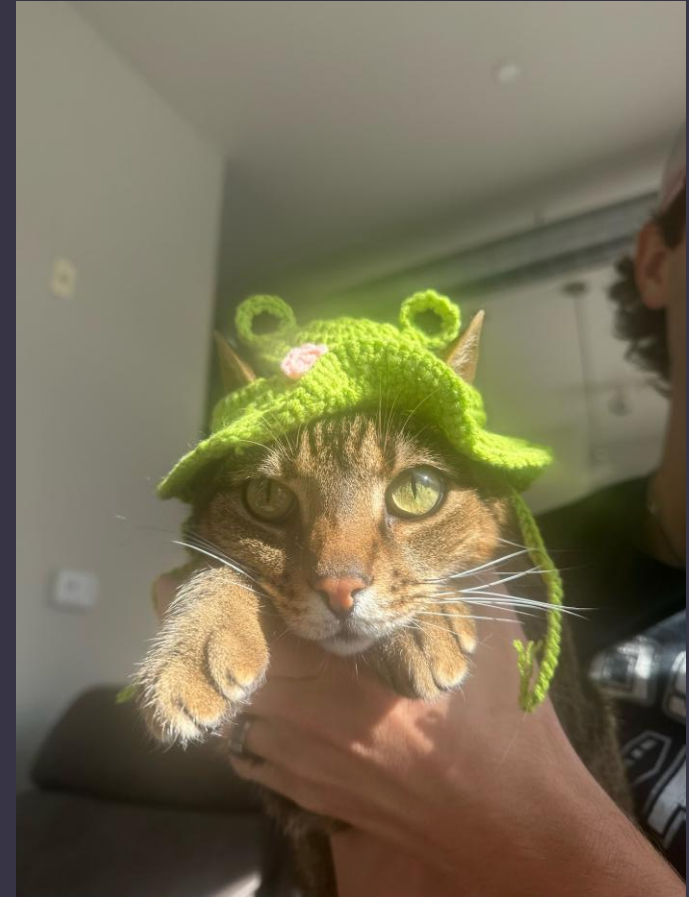
Sample Inaccuracy: Species ID



Lobster

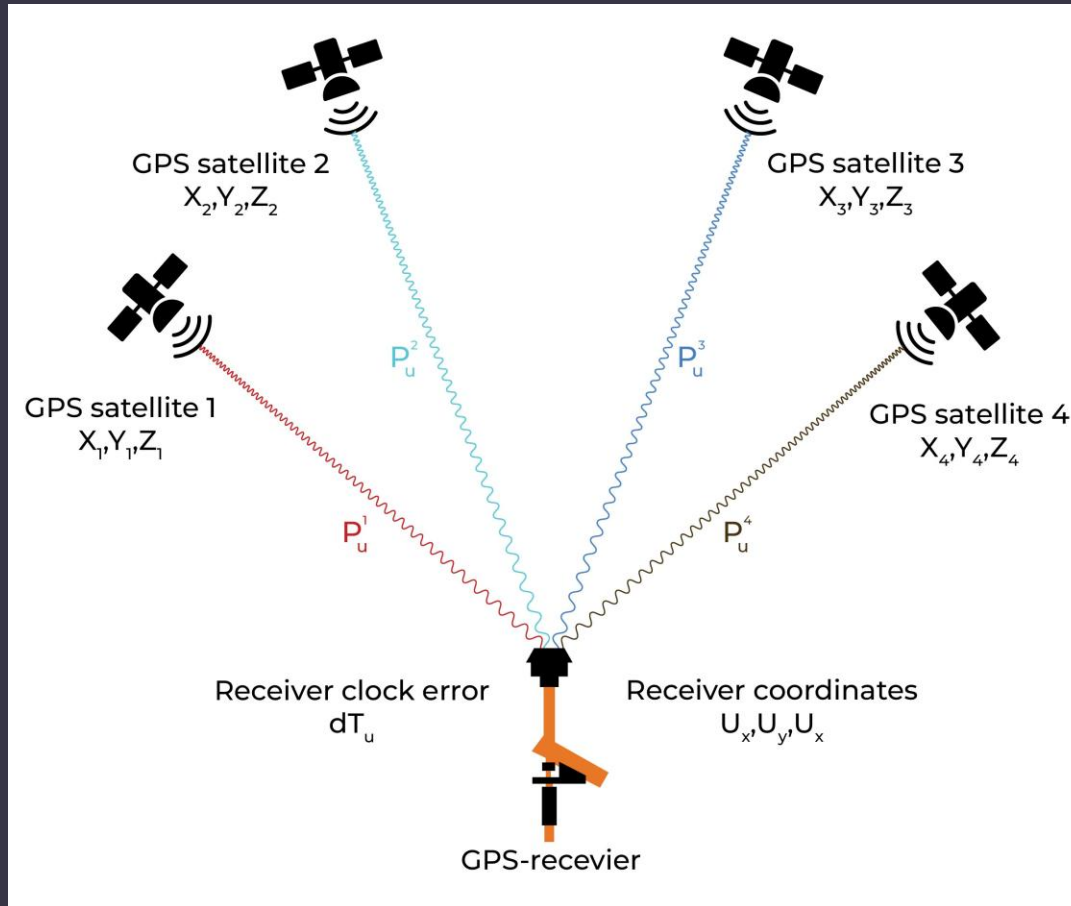


Turkey



Frog (photo cred: Gracie)

Sample Inaccuracy: Spatial Inaccuracy



What can I do about record inaccuracy?



Solanum acaule
Wild potato found in the
Andes

What can I do about record inaccuracy?

Visually inspect:

- Records that are far out of the range of other records
- Records that don't match administrative data
- Records associated with collections facilities
- Records in improbable locations
 - Physiologically, in middle of ocean, Antarctica



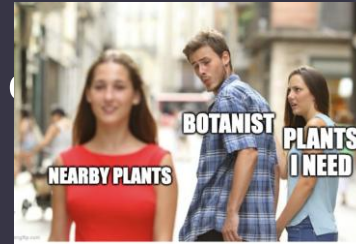
Solanum acaule
Wild potato found in the
Andes



Violations

#1: Collector Bias

- People collect where it is easy to collect (roadside, near infrastructure, in safe regions, etc.)
- Solution: Spatial thinning AND background point generation (more in a sec)

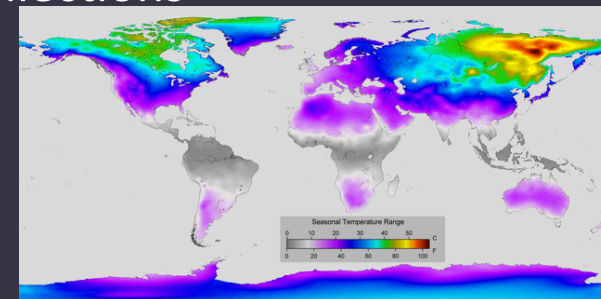


#2: Spatial Autocorrelation

- Species live in populations; members *aren't* randomly distributed. Environmental variables are also autocorrelated
- Solution: Systematic or stratified sub-sampling to maximize nearest neighbor distance

#3: Temporal Variation

- Individuals sampled more than once, or one particular area has multiple collections
- Solution: Remove spatial duplicates in your dataset (all but one, of course)
- Some organisms vary in abiotic requirements by season
- Solution: Filter and train models by month/season/time-span



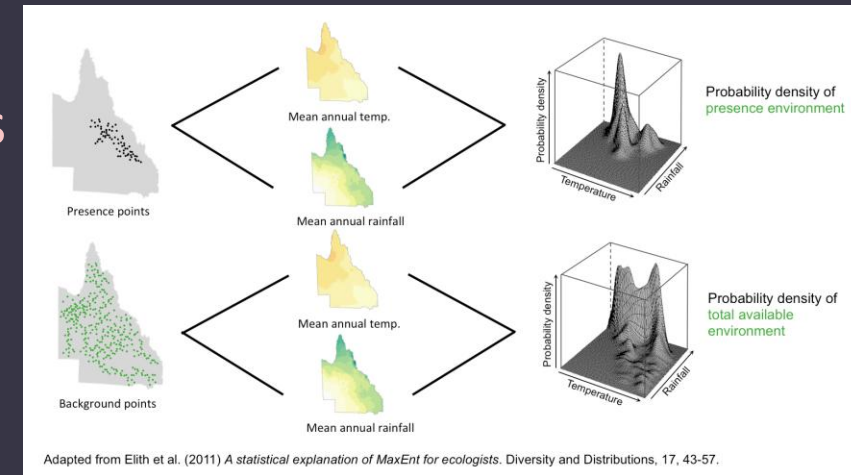
Background Data

Absence/Pseudo-Absence/Background: Where a species is *not* known to occur

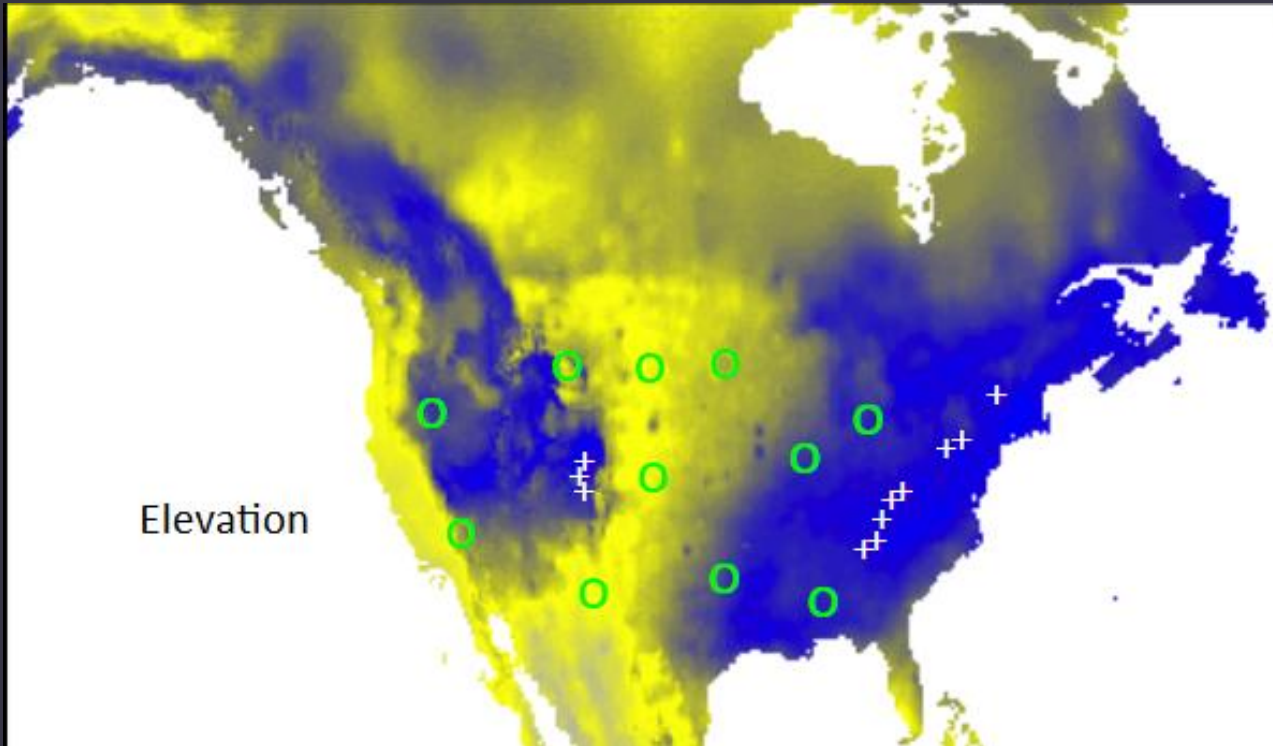
- Unless used surveys include data about where species were searched for and not found, this data is just a guess

Serves two functions:

1. Provide contrast to the presence/predictor correlation
2. Contributes to model evaluation: how well does the model predict presences compared to random data?



Background Data to Gauge Correlation



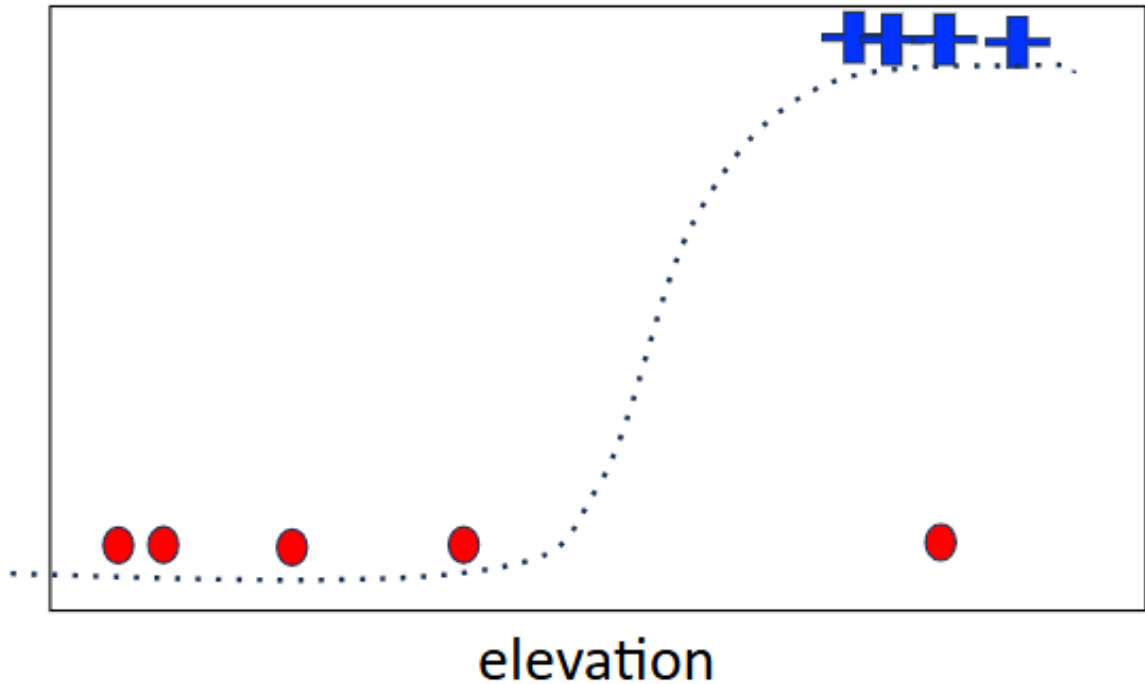
Crosses denote presence, circles are background points

How well does our presence data correlate with elevation?

It depends on the variation of elevation in the landscape we are looking at

Background data samples that variation

Background Data to Gauge Correlation



How well does our presence data correlate with elevation?

It depends on the variation of elevation in the landscape we are looking at

Background data samples that variation

Presence data correlates with elevation

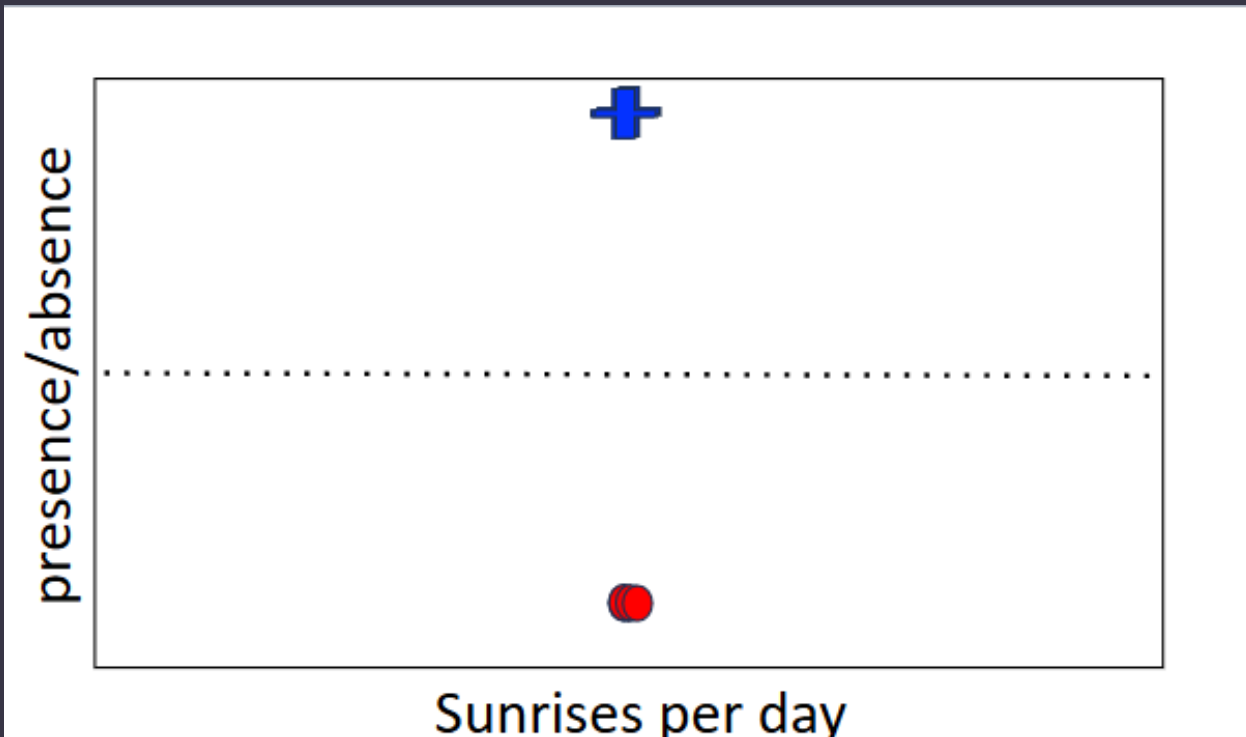
Background Data to Gauge Correlation



Crosses denote presence, circles are background points

How about with average number of sunrises per day?

Absence Data to Gauge Correlation



Context is so common, with so little variation, as to be an essentially meaningless predictor

- The next talk will touch on variable selection

Presence data does not correlate with sunrise frequency

History of Methods

Early Approach (1990-2006ish)

Can we distinguish habitats in which species occur from ones in which they don't? (Absence)

- Identify where species aren't to some degree of certainty

Newer Approach (2006ish-Now)

Can we distinguish habitats in which species are known to occur from habitats in which they are not known to occur (Background/Pseudo-absence)

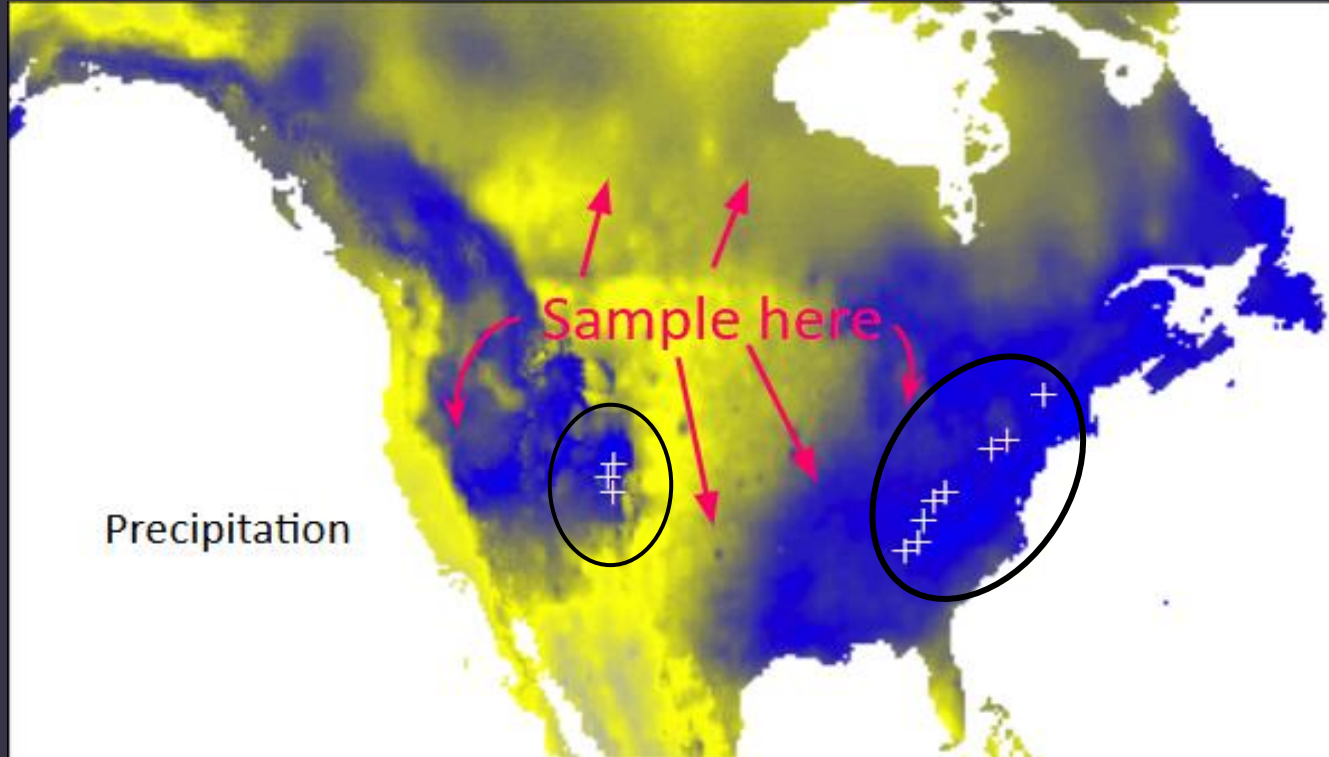
- Occurrence VS Random Sample
- More biologically realistic and conservative



Early Approach: Method #1

Geographically stratified

Define a species range (buffer around occurrence points) and randomly sample outside

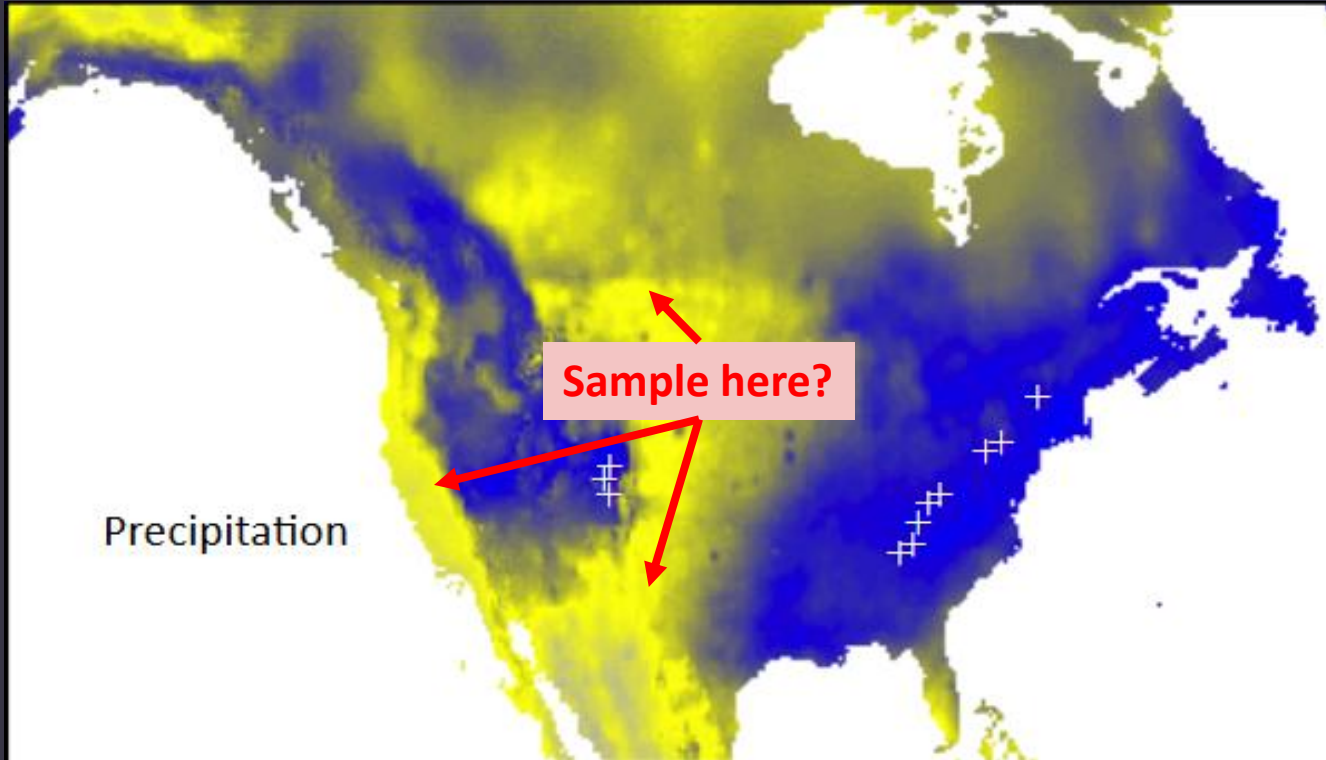


- Buffer is arbitrary
- How large of an area beyond the buffer do we sample (extent)?
 - A few degrees?
 - Within expected dispersal range?
 - Continent?
 - Universe?

Early Approach: Method #2

Environmentally stratified

Sample only in areas in which environmental conditions are different



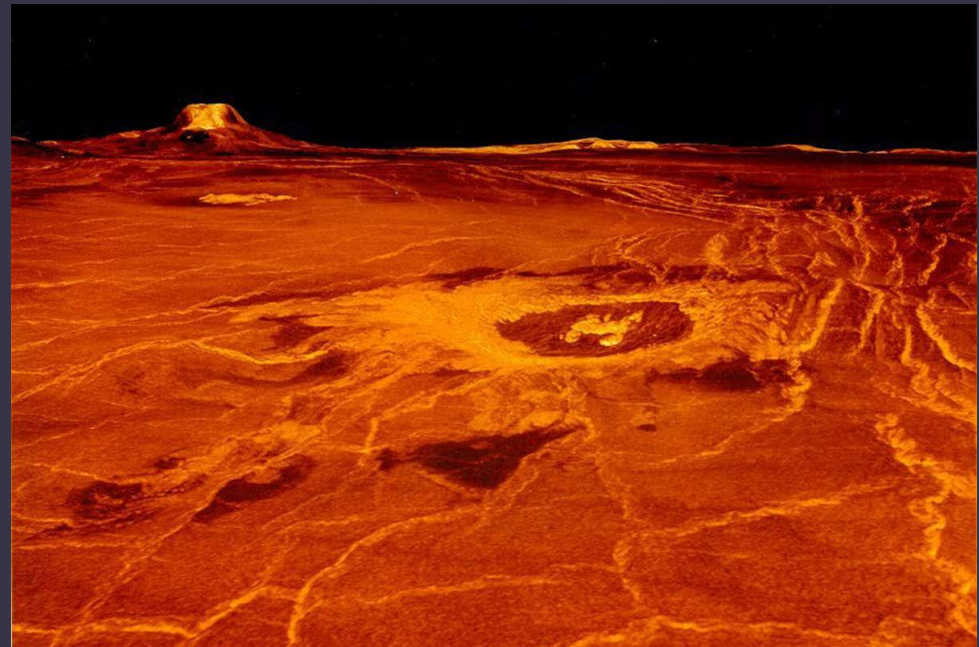
- Which conditions?
- What is considered different?
- How large of an area beyond the buffer do we sample (extent)?
 - A few degrees?
 - Within expected dispersal range?
 - Continent?
 - Universe?

An Aside: Scale Does Matter for This

We can 100% say that this species does not occur on any planet other in our solar system than Earth...



...But how useful is a model that can distinguish terrestrial habitats from those on Venus?



Extent Selection

Picking an extent that is too large may over inflate discriminatory performance metrics

Picking an extent that is too small may mean you aren't exposing your models to enough variation in the environmental landscape

Okay, so what should we do?

1. Some argue it should be determined from an ecological theory standpoint (Elith et al. 2011)
2. Some argue for the use of statistical metrics to bound background point generation
3. ALL agree that it influences the outcome of your model (at least to some extent)



Minimizing effects of methodological decisions on interpretation and prediction in species distribution studies: An example with background selection

Catherine S. Jarnevich^{a,*}, Marian Talbert^b, Jeffery Morissette^b, Cameron Aldridge^c, Cynthia S. Brown^d, Sunil Kumar^e, Daniel Manier^e, Colin Talbert^{b,f}, Tracy Holcombe^e

Determining the minimal background area for species distribution models: MinBAR package

Xavier Rotllan-Puig^{a,*}, Anna Traveset^b

Extent Selection

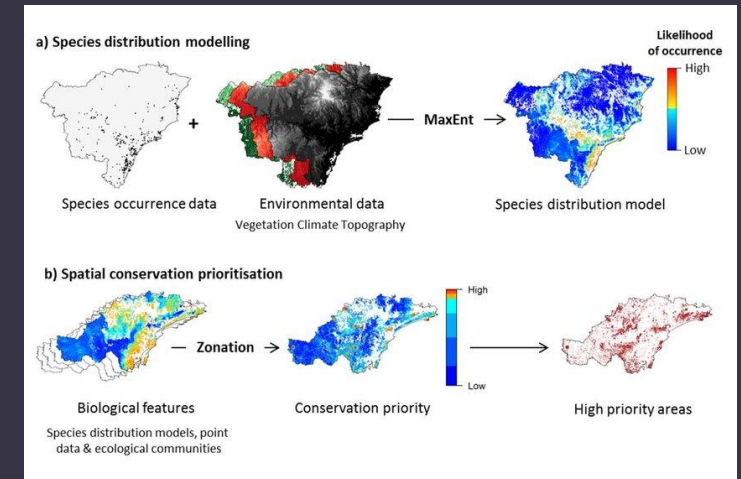
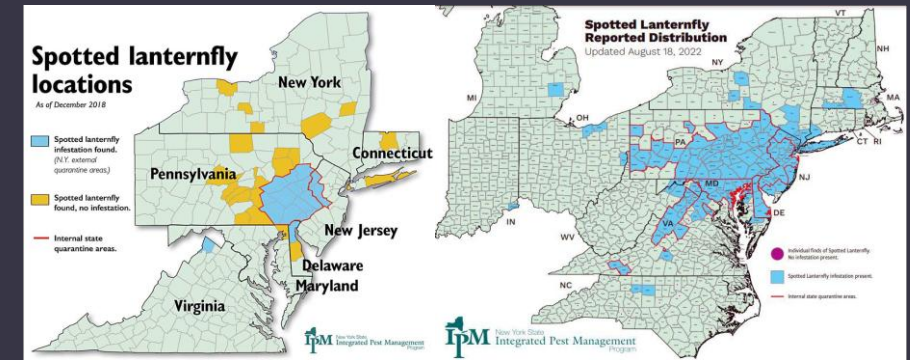
As always seems to be the case in niche modeling, most arguments are valid under some circumstances and the choice depends on 1) the study system, and 2) the study question

Invasion Risk

- When looking at invasion risk we often want to look at how the species will perform in novel places **AND** times
 - Extent will need to be larger, such that we are projecting onto an area where environmental variation has been sampled

Species on the Decline

- In the case of conservation, identify where it is within the current range that suitability will decline **OR** remain stable/increase
 - Such that funds/efforts can be focused on areas that will remain habitable
 - In this case, we do not need to be extending our “buffer” out too far



Refined Approach

Treat “absences” as background/pseudo-absence points

- Models are designed to distinguish presence points from random points

Background sample should have the same collector bias as occurrence sample

- Phillips et al. 2009, Sample selection bias and presence-only distribution models: implications for background and pseudo-absence data

How do we implement this?

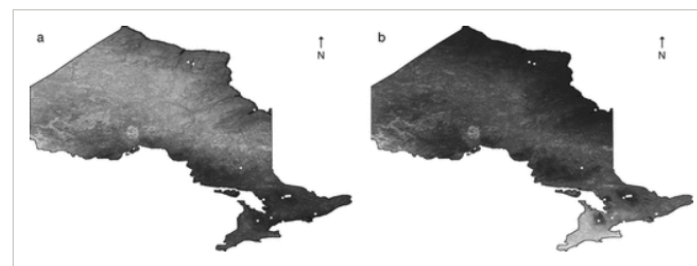
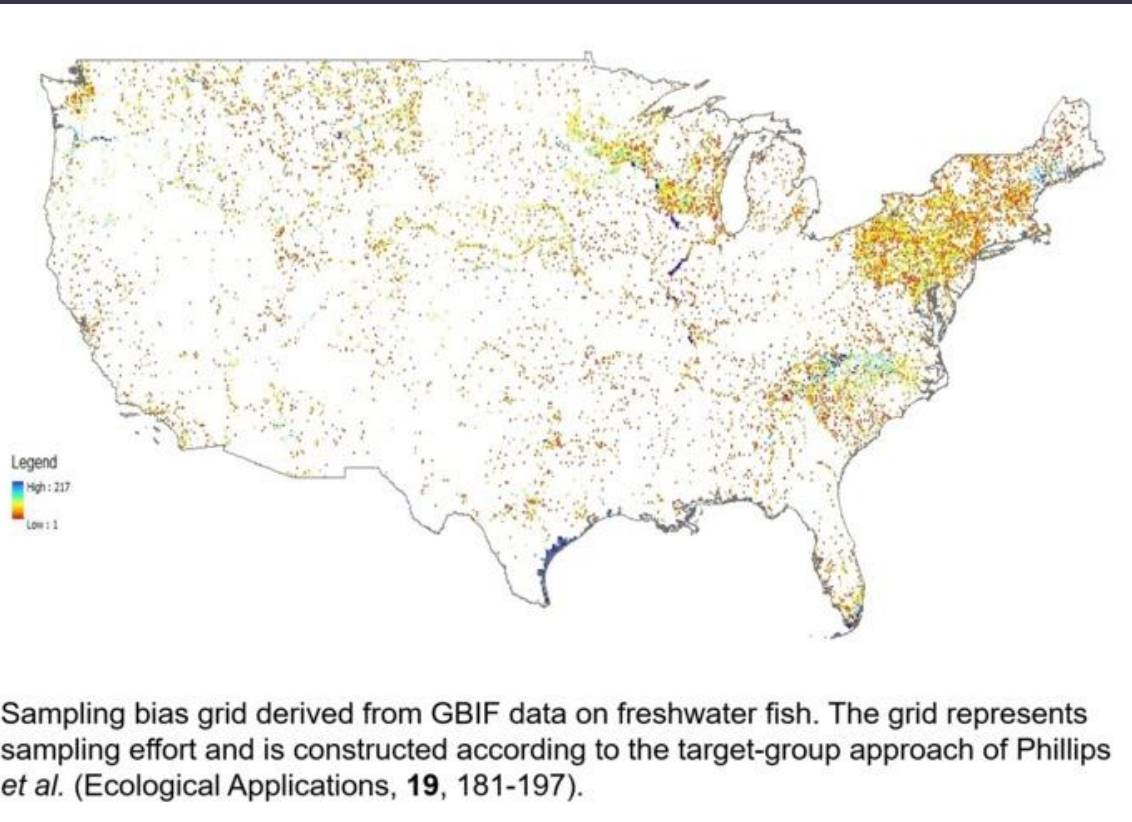


Figure 7

[Open in figure viewer](#) | [Download PowerPoint](#)

Maxent predictions in Ontario, Canada for the Golden-crowned Kinglet, a widely distributed generalist species, created (a) without and (b) with use of target-group background. Dark shades indicate stronger prediction, while white or black dots are presence sites used in training. Without target-group background, the prediction is similar to the model of sampling effort (Fig. 3). Target-group background results in stronger prediction in less-sampled areas, reducing dependency of sampling effort.

Refined Approach: Method #1



1. Replicate sample bias by using occurrence points from species with similar collection histories as your background/pseudo-absence data

- However, issue of scale remains

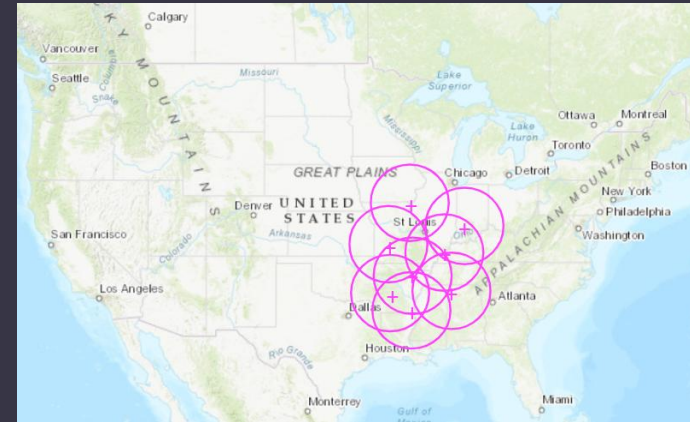
Refined Approach: Method #2

2. Sample from within the range of occurrences

- Nearest neighbor, within a defined buffer, etc.
- Can also do a “combined” manner of sampling

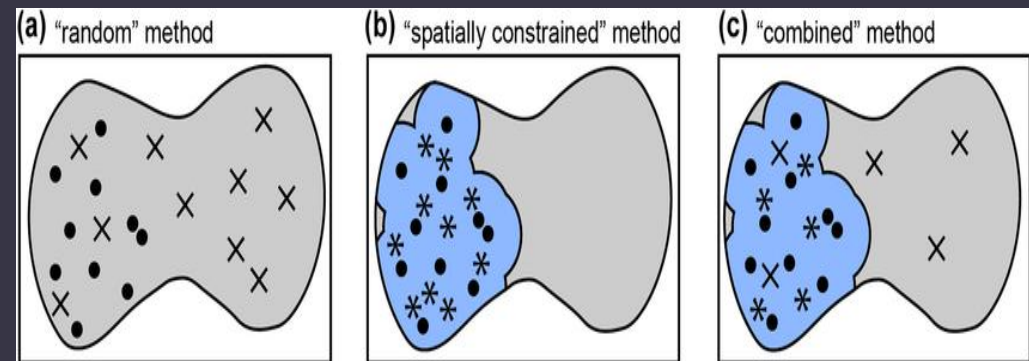
A bit more challenging for the model

- Has to discriminate occurrence points from random points in an area with very similar environmental conditions



Scale issue remains

- How big do you make the buffer?
- If sampling in and out, what weighting scheme do you use?
- As above, how far outside of the buffer should we sample (extent)?



In Review

Raw and readily available occurrence data does not conform to our standards of a perfect sample

This leads to violations that need to be addressed either through:

- Cleaning our occurrence records in specific ways
- Generating our background data in certain ways

Walked through the history of background point generation

Discussed the most common current methods to do that

Talked about how the study extent influences model projection and accuracy metrics, and considerations for how we can deal with that

Next: Predictor variables: how to pick and deal with them

